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**04 April 2026**
**Total Citations:** 160
**Total Downloads:** 1788

**Published:** 01 August 2012
**Accepted:** 01 July 2011
**Received:** 01 February 2011

**Citation in BibTeX format**

# Algorithm 923: Efficient Numerical Computation of the Pfaffian for Dense and Banded Skew-Symmetric Matrices

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Computing the Pfaffian of a skew-symmetric matrix is a problem that arises in various fields of physics. Both computing the Pfaffian and a related problem, computing the canonical form of a skew-symmetric matrix under unitary congruence, can be solved easily once the skew-symmetric matrix has been reduced to skew-symmetric tridiagonal form. We develop efficient numerical methods for computing this tridiagonal form based on Gaussian elimination, using a skew-symmetric, blocked form of the Parlett-Reid algorithm, or based on unitary transformations, using block Householder transformations and Givens rotations, that are applicable to dense and banded matrices, respectively. We also give a complete and fully optimized implementation of these algorithms in Fortran (including a C interface), and also provide Python, Matlab and Mathematica implementations for convenience. Finally, we apply these methods to compute the topological charge of a class D nanowire, and show numerically the equivalence of definitions based on the Hamiltonian and the scattering matrix.

Categories and Subject Descriptors: G.1.3 [**Numerical Linear Algebra**]; G.4 [**Mathematical Software**]; J.2 [**Physical Sciences and Engineering**] Physics

General Terms: Algorithms, Performance

Additional Key Words and Phrases: Pfaffian, skew-symmetric matrix, canonical form, unitary congruence, topological charge

**ACM Reference Format:**
Wimmer, M. 2012. Algorithm 923: Efficient numerical computation of the Pfaffian for dense and banded skew-symmetric matrices. ACM Trans. Math. Softw. 38, 4, Article 30 (August 2012), 17 pages.
DOI = 10.1145/2331130.2331138 http://doi.acm.org/10.1145/2331130.2331138

## 1. INTRODUCTION

### 1.1. Pfaffians and reduction to tridiagonal form

A real or complex matrix $A$ is called *skew-symmetric* (or *antisymmetric*), if $A = -A^T$, where $^T$ denotes the transpose. The determinant $\det(A)$ of such a skew-symmetric matrix is the square of a polynomial in the matrix entries, the *Pfaffian* $\mathrm{Pf}(A)$:

$$\det(A) = \mathrm{Pf}(A)^2 . \tag{1}$$

In other words, the Pfaffian of a skew-symmetric matrix is a unique choice for the sign of the root of the determinant:

$$\mathrm{Pf}(A) = \pm\sqrt{\det(A)}. \tag{2}$$

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© 2012 ACM 0098-3500/2012/08-ART30 $15.00
DOI 10.1145/2331130.2331138 http://doi.acm.org/10.1145/2331130.2331138

Pfaffians arise in various fields of physics, such as in the definition of topological charges [Fu and Kane 2007; Fulga et al. 2011; Kitaev 2001], electronic structure quantum Monte Carlo [Bajdich et al. 2008; Bajdich and Mitas 2009], the two-dimensional Ising spin glass [Thomas and Middleton 2009], or in the definition of a trial wave function for the $\nu = 5/2$ fractional quantum Hall state [Moore and Read 1991]. It also arises naturally from Gaussian Grassmann integration, and as such finds applications for example in quantum chaos [Haake 2004] or lattice quantum field theory [Montvay 1996].

The Pfaffian for a $2n \times 2n$ skew-symmetric matrix is defined as

$$\mathrm{Pf}(A) = \frac{1}{2^n n!} \sum_{\sigma \in S_{2n}} \mathrm{sgn}(\sigma) \prod_i^n a_{\sigma(2i-1),\sigma(2i)}, \tag{3}$$

where $S_{2n}$ is the group of permutations of sets with $2n$ elements. The Pfaffian of an odd-dimensional skew-symmetric matrix is defined to be zero, as in this case also $\det(A) = 0$ ($\det(A) = \det(A^T) = \det(-A) = (-1)^{2n-1}\det(A)$). While Equation (3) can be used to compute the Pfaffian directly for small matrices, its computational cost $\mathcal{O}(n!)$ is prohibitively expensive for larger matrices.

Analogous to the numeric computation of the determinant, a promising strategy is thus to find a transformation of the original matrix into a form that allows an easier evaluation of the Pfaffian. Particularly useful in this context is the recursive definition of the Pfaffian,

$$\mathrm{Pf}(A) = \sum_{i=2}^{2n} (-1)^i a_{1i} \mathrm{Pf}(A_{1i}), \tag{4}$$

where $A_{1i}$ is the matrix A without the rows and columns 1 and $i$. (Note that the Pfaffian of a $0 \times 0$ matrix is defined as 1). Further, for an arbitrary $2n \times 2n$ real or complex matrix $B$,

$$\mathrm{Pf}(BAB^T) = \det(B)\mathrm{Pf}(A). \tag{5}$$

From the recursive definition of the Pfaffian (4) it is obvious that the Pfaffian of a $2n \times 2n$ skew-symmetric tridiagonal matrix

$$T = \begin{pmatrix} 0 & a_1 & & & & & \\ -a_1 & 0 & b_1 & & & & \\ & -b_1 & 0 & a_2 & & & \\ & & -a_2 & \ddots & \ddots & & \\ & & & \ddots & 0 & b_{n-1} & \\ & & & & -b_{n-1} & 0 & a_n \\ & & & & & -a_n & 0 \end{pmatrix} \tag{6}$$

is given as

$$\mathrm{Pf}(T) = \prod_{i=1}^n a_i. \tag{7}$$

Furthermore, a closer inspection of Equation (4) shows that also a matrix that has only a partial tridiagonal form with $t_{ij} = t_{ji} = 0$ only for odd $i$ and $j > i+1$ (i.e., a matrix that would be tridiagonal, if every *even* row and column would be removed),

$$\tilde{T} = \begin{pmatrix} 0 & a_1 & & & & & \\ -a_1 & 0 & t_{23} & t_{24} & t_{25} & \cdots & \\ & -t_{23} & 0 & a_2 & & & \\ & -t_{24} & -a_2 & \ddots & \ddots & & \\ & -t_{25} & & \ddots & 0 & t_{2n-2,2n-1} & t_{2n-2,2n} \\ & \vdots & & & -t_{2n-2,2n-1} & 0 & a_n \\ & & & & -t_{2n-2,2n} & -a_n & 0 \end{pmatrix} \tag{8}$$

allows for an easy evaluation of the Pfaffian since $\mathrm{Pf}(\tilde{T}) = \mathrm{Pf}(T)$, evident from Equation (4).

Our goal is therefore to find for a skew-symmetric matrix $A$ a suitable transformation $B$ such that

$$A = BTB^T \tag{9}$$

with $T$ tridiagonal or tridiagonal in every odd row and column.

It has been known for a while that the Pfaffian of a skew-symmetric $n \times n$ matrix $A$ can be computed in $\mathcal{O}(n^3)$ time, using a skew-symmetric form of Gaussian elimination (adding multiples of rows and columns in a symmetric fashion) [Bajdich and Mitas 2009; Galbiati and Maffioli 1994; Krauth 2006; Rote 2001]. Such an skew-symmetric Gaussian elimination computes a factorization of the matrix in the form (9) with $B = PL$ where $P$ is a permutation matrix and $L$ a unit lower triangular matrix. For brevity, we will refer to this type of decomposition as $LTL^T$ decomposition. Gaussian elimination requires pivoting for numerical stability, hence the need for the permutation $P$. In the following, we will formulate this approach in a way that allows for an efficient computer implementation.

Another Gaussian-based elimination technique is the $LDL^T$ decomposition where $A$ is reduced to $D$, a matrix with only skew-symmetric $2 \times 2$-blocks on the diagonal [Bunch 1982; Higham 2002]. This approach has also been suggested for computing the Pfaffian recently [Rubow and Wolff 2011; Thomas and Middleton 2009]. We will not persue this approach here, but show that the $LTL^T$ decomposition allows for computing the Pfaffian in the same number of operations and can be formulated more easily to use level-3 matrix operations.

As an alternative to the Gaussian elimination based techniques, we also develop algorithms using unitary (orthogonal in the real case) transformations that are also known to allow for a stable numerical computation in $\mathcal{O}(n^3)$ for dense matrices. This approach does not require pivoting for numerical stability and can more easily make use of the bandedness of a matrix. We will describe how to compute such a unitary matrix $Q$ in order to tridiagonalize (either fully or partially) $A$,

$$A = QTQ^T, \tag{10}$$

or equivalently $T = Q^\dagger A Q^*$, where $^\dagger$ denotes the Hermitian conjugate and $^*$ complex conjugation. Note that such a *unitary congruence transformation* is for the complex case quite different from the usual unitary similarity transformations one usually encounters, which are of the form $A = QTQ^\dagger$. In the real case, the transformation reduces to the usual orthogonal similarity transformation.

### 1.2. Tridiagonalization and the Canonical Form of Skew-Symmetric Matrices

Apart from allowing for an efficient computation of the Pfaffian, the tridiagonal form of a skew-symmetric matrix under unitary congruence is also relevant for computing the canonical form of this matrix.

A skew-symmetric matrix has a particularly simple canonical form under a unitary congruence transformation. For every skew-symmetric matrix $A$ there exists a unitary matrix $U$ such that [Hua 1944; Stander and Wiegman 1960]

$$A = U\Xi U^T, \quad \text{where } \Xi = \Sigma_1 \oplus \Sigma_2 \oplus \cdots \oplus \Sigma_k \oplus 0 \oplus \cdots \oplus 0, \tag{11}$$

where $\operatorname{rank}(A) = 2k$, $\oplus$ denotes the direct sum, and

$$\Sigma_j = \begin{pmatrix} 0 & \sigma_j \\ -\sigma_j & 0 \end{pmatrix}, \qquad \sigma_j > 0. \tag{12}$$

This canonical form has been used in the physics context for example to prove the Kramer's degeneracy of transmission eigenvalues [Bardarson 2008] and the degeneracy of Andreev reflection eigenvalues [Béri 2009].

The problem of computing the canonical form of an even-dimensional skew-symmetric tridiagonal matrix has been discussed in [Golub and Van Loan 1996; Hastings and Loring 2011; Ward and Gray 1978b], the reduction of the problem with on odd-dimensional matrix to the even-dimensional case in [Ward and Gray 1978b]. For a $2n \times 2n$ skew-symmetric tridiagonal matrix as defined Equation (6), the values of $\sigma_i$, $i = 1..k$ are given by the $k$ nonzero singular values of the bidiagonal matrix

$$J = \begin{pmatrix} a_1 & -b_1 & & & \\ & a_2 & -b_2 & & \\ & & \ddots & \ddots & \\ & & & a_{n-1} & -b_{n-1} \\ & & & & a_n \end{pmatrix}. \tag{13}$$

For completeness, we give details and a self-contained derivation in Appendix 1.

The canonical form of a skew-symmetric matrix under unitary congruence is also connected to certain eigenvalue problems: In the real case, the eigenvalues of $A$ are given by $\pm i\sigma_j$. In the complex case, the matrix $A^*A = -A^\dagger A$ has doubly degenerate eigenvalues $\sigma_j^2$.

### 1.3. Skew-Symmetric Tridiagonalization and Existing Approaches

Both the computation of the Pfaffian and of the canonical form are ultimately connected to the problem of tridiagonalizing a skew-symmetric matrix. Here we give an overview of existing solutions (with implementations) that could be used to solve parts of the problem, and discuss the need for a new comprehensive implementation.

For real skew-symmetric matrices, the unitary congruence transformation reduces to an ordinary orthogonal similarity transformation and hence established decompositions can be used [Golub and Van Loan 1996]: The Hessenberg decomposition of a skew-symmetric matrix reduces to tridiagonal form (6), and the real Schur decomposition to the canonical form (11) (implemented, for example in LAPACK [Anderson et al. 1999]). However, none of these decompositions make use of the structure of the problem which would be desirable for precision and speed, nor can they be used for complex skew-symmetric matrices.

Ward and Gray have developed and implemented algorithms to compute the tridiagonal form and the eigenvalues (and as an intermediate step, the canonical form) of a

real dense, skew-symmetric matrix, making use of the structure of the problem [Ward and Gray 1978a]. A complex version is however not available.

The accompanying Matlab code to Higham [2002] contains a skew-symmetric $LDL^T$ decomposition that can be used to compute Pfaffians, but according to the authors is not designed for efficiency.

Very recently, González-Ballestero et al. [2011] have developed a library for the numerical computation of the Pfaffian of a dense skew-symmetric matrix, but do not give access to the transformation matrix (e.g. needed for computing the canonical form). They present algorithms based on a $LDL^T$ decomposition (called Aitken block diagonalization in González-Ballestero et al. [2011]) and on Householder tridiagonalization. However, their approach does not make use of the full symmetry of the problem.

None of the existing approaches (with the exception of LAPACK that does not exploit the skew-symmetry of the problem) makes use of block algorithms that are rich in level-3 operations and desirable for a more favorable memory access pattern. In the following, we will show that such block algorithms can give rise to a considerable increase in speed.

Moreover, none these approaches makes use of the sparsity of a banded matrix, a structure that however often arises in practice. In the following, we will also consider this case in particular.

The goal of this work is thus to develop and implement algorithms for tridiagonalizing a real or complex skew-symmetric matrix, making use of the skew-symmetry and possibly the bandedness of the matrix.

### 1.4. Outline

The remainder of the article is organized as follows. In Section 2 we discuss algorithms to tridiagonalize a dense or banded skew-symmetric matrix using Gaussian elimination, Householder reflections and Givens rotations. Further, in Section 3 we discuss the details of our implementation, and present benchmarks and an exemplary application in Section 4. In the appendix, we give a self-contained derivation on the computation of the canonical form of a tridiagonal, skew-symmetric matrix. Moreover, we discuss a version of our tridiagonalization algorithm for dense matrices making use of block Householder reflections.

## 2. SKEW-SYMMETRIC NUMERICAL TRIDIAGONALIZATION

### 2.1. Statement of the Problem

Summarizing the previous discussion, for a given skew-symmetric $n \times n$ matrix $A$ we seek a (invertible) transformation $B$ such that $A = BTB^T$ with $T$ in tridiagonal form (or in partial tridiagonal form). We will consider first an algorithm for dense matrices based on Gauss transformations requiring pivoting. Then we focus on algorithms bases on unitary transformations where we consider both dense and banded matrices. The discussion is presented for the case of complex matrices, but it carries over to the real case unchanged.

### 2.2. $LTL^T$ Decomposition of Dense Matrices Using the Parlett-Reid Algorithm

For symmetric or Hermitian matrices there exist efficient algorithms to compute a $LTL^T$ or $LDL^T$ decomposition (for an overview, see Golub and Van Loan [1996]). It has been shown by Bunch that those decompositions can in principle also be generalized and computed stably for skew-symmetric matrices [Bunch 1982]. In the following we reformulate the algorithm for the $LTL^T$ decomposition of a symmetric matrix due to Parlett and Reid [1970] such that it is suitable for skew-symmetric matrices. The Parlett-Reid algorithm is usually not the method of choice in the symmetric

case, as there are more efficient alternatives [Aasen 1971; Bunch and Kaufman 1977; Rozložník et al. 2011]. However, as we will discuss further, the Parlett-Reid algorithm can be used to compute the *Pfaffian* just as effectively.

A $n \times n$ matrix of the form

$$M_k = E_n - \boldsymbol{\alpha}_k(\mathbf{e}_k^{(n)})^T \tag{14}$$

where $E_n$ is the $n \times n$ identity matrix and $\mathbf{e}_k^{(n)}$ the $k$-th unit vector in $\mathbb{C}^n$, is called a *Gauss transformation* if the first $k$ entries of $\boldsymbol{\alpha}_k$ are zero. Given a vector $\mathbf{x} = (x_1 \dots x_n)^T$ and taking $\boldsymbol{\alpha}_k = (0 \dots 0 \;\; x_{k+1}/x_k \dots x_n/x_k)^T$, $M_k$ can be used to eliminate the entries $k+1 \dots n$ in $\mathbf{x}$, $M_k\mathbf{x} = (x_1 \dots x_k \; 0 \dots 0)$, provided that $x_k \neq 0$.

A Gauss transformation can thus be used to zero the entries in a column or row of $A$ below a chosen point $k$. In order to avoid divisions by a small number or zero, a permutation $P_k$ interchanging entry $k$ with another nonzero, typically the largest entry in $k+1 \dots n$ is performed. The numerical stability of this pivoting strategy is discussed in Bunch [1982].

Hence, a series of Gauss transformations and permutations can be used to tridiagonalize a skew-symmetric matrix $A$. To demonstrate the mechanism, assume that after applying $k-1$ Gauss transformations and permutations, the matrix $A^{(k-1)} = M_k P_k \dots M_2 P_2 \, A \, P_2^T M_2^T \dots P_k^T M_k^T$ is already in tridiagonal form in the first $k-1$ columns and rows and hence has the form

$$A^{(k-1)} = \begin{pmatrix} A_{11} & A_{12} & 0 \\ A_{21} & 0 & A_{23} \\ 0 & A_{32} & A_{33} \end{pmatrix} \begin{matrix} k-1 \\ 1 \\ n-k \end{matrix} \tag{15}$$

with $A_{11} \in \mathbb{C}^{k-1 \times k-1}$, $A_{21} \in \mathbb{C}^{1 \times k-1}$, $A_{32} \in \mathbb{C}^{n-k \times 1}$, $A_{33} \in \mathbb{C}^{n-k \times n-k}$, and $A_{12} = -A_{21}^T$, $A_{23} = -A_{32}^T$ (transformations of the form $BAB^T$ maintain skew-symmetry). Now choose a permutation matrix $\tilde{P}_{k+1}$ such that the maximal entry in $A_{32} = (a_{k+1} \dots a_n)^T$ is permuted to the top, i.e. $\tilde{P}_{k+1}A_{32} = (\tilde{a}_{k+1} \dots \tilde{a}_n)^T$ where $|\tilde{a}_{k+1}| = \max(|a_{k+1}|, \dots |a_n|)$. If the maximal element at this step is zero, $A_{32} = 0$ and $A^{(k-1)}$ is already tridiagonal in the first $k$ columns and we set $M_{k+1} = P_{k+1} = E_n$. Otherwise, we take $P_{k+1} = \mathrm{diag}(E_k, \tilde{P}_{k+1})$ and $M_{k+1} = \mathrm{diag}(E_k, \tilde{M}_{k+1})$ with $\tilde{M}_{k+1} = E_{n-k} - \boldsymbol{\alpha}_{k+1}(\mathbf{e}_1^{(n-k)})^T$ with $\boldsymbol{\alpha}_{k+1} = (0 \; \tilde{a}_{k+2}/\tilde{a}_{k+1} \dots \tilde{a}_n/\tilde{a}_{k+1})^T$. Then we obtain

$$A^{(k)} = M_{k+1}P_{k+1}A^{(k-1)}P_{k+1}^T M_{k+1}^T = \begin{pmatrix} A_{11} & A_{12} & 0 \\ A_{21} & 0 & A_{23}\tilde{P}_{k+1}^T\tilde{M}_{k+1}^T \\ 0 & \tilde{M}_{k+1}\tilde{P}_{k+1}A_{32} & \tilde{M}_{k+1}\tilde{P}_{k+1}A_{33}\tilde{P}_{k+1}^T\tilde{M}_{k+1}^T \end{pmatrix}. \tag{16}$$

Then $\tilde{M}_{k+1}\tilde{P}_{k+1}A_{32} \propto \mathbf{e}_1^{(n-k)}$ and $A^{(k)}$ is tridiagonal in its first $k$ rows and columns (here $\propto$ denotes "proportional to"). Defining $\mathbf{w} = \tilde{P}_{k+1}A_{33}\tilde{P}_{k+1}^T\mathbf{e}_1^{(n-k)}$ we find

$$\tilde{M}_{k+1}\tilde{P}_{k+1}A_{33}\tilde{P}_{k+1}^T\tilde{M}_{k+1}^T = \tilde{P}_{k+1}A_{33}\tilde{P}_{k+1}^T + \boldsymbol{\alpha}_{k+1}\mathbf{w}^T - \mathbf{w}\boldsymbol{\alpha}_{k+1}^T. \tag{17}$$

The cross-term $(\mathbf{e}_1^{(n-k)})^T\tilde{P}_{k+1}A_{33}\tilde{P}_{k+1}^T\mathbf{e}_1^{(n-k)}$ vanishes due to the skew-symmetry of $A_{33}$. Note that in this skew-symmetric outer product update, the matrix $\tilde{P}_{k+1}A_{33}\tilde{P}_{k+1}^T$ remains actually unchanged in the first column and row due to the structure of $\boldsymbol{\alpha}_{k+1}$ and $\mathbf{w}$. The outer product update is dominating the computational cost of each step

$$\begin{pmatrix} 0 & \times & & & & \\ \times & 0 & \times & \boxdot & \boxdot & \boxdot \\ & & \times & 0 & \times & \times & \times \\ & & \boxdot & \times & 0 & \times & \times \\ & & \boxdot & \times & \times & 0 & \times \\ & & \boxdot & \times & \times & \times & 0 \end{pmatrix}$$

Fig. 1. Example of the structure induced by applying a Gauss or Householder transformation from left and right. $\times$ represents nonzero entries and $\boxdot$ those entries that are zeroed by the Gauss or Householder transformation. In this example the first column and row have already been reduced, and the transformation is applied to $A^{(1)}$ from the left and the right. The parts of the matrix that are changed in the process are marked with a frame.

and can be computed in $2(n-k)^2$ flops, if the symmetry is fully accounted for. Figure 1 shows the structure of a tridiagonalization step schematically for a particular example.

After $n-2$ steps, the decomposition can be written as

$$PAP^T = LTL^T \tag{18}$$

with permutation $P = P_{n-1} \dots P_2$, skew-symmetric tridiagonal $T = M_{n-1}P_{n-1} \dots M_2 P_2 A P_2^T M_2^T \dots P_{n-1}^T M_{n-1}^T$, and lower unit triangular matrix

$$L = (M_{n-1}P_{n-1} \dots M_2 P_2 P^T)^{-1}. \tag{19}$$

As in the symmetric Parlett-Reid algorithm, the first column of $L$ is $\mathbf{e}_1^{(n)}$, and the $k$-th column below the diagonal a permuted version of $\boldsymbol{\alpha}_k$.

The computation of the updated matrix, Equation 17, is a level-2 matrix operation. It is possible to regroup these updates in a way that allows to operate with level-3 matrix operations that have a more favorable memory access pattern. The details of this block version of the Parlett-Reid algorithm are given in Appendix 2.

The full skew-symmetric $LTL^T$ decomposition can be computed in $2n^3/3$ flops. It is however readily generalized to compute only a partial tridiagonal form as in Equation 8 by skipping every other row and column elimination. This partial $LTL^T$ decomposition can thus be computed in $n^3/3$ flops. Since $\det(L) = 1$ and $\det(P)$ can be computed in $n$ steps, computing the Pfaffian of a skew-symmetric matrix with the Parlett-Reid algorithm thus requires a total of $n^3/3$ flops. This is a factor of 10 less than the unsymmetric Hessenberg decomposition.

For computing a full tridiagonalization, the Parlett-Reid algorithm requires twice as many flops as other approaches: Aasen's algorithm [Aasen 1971] computes a (complete) $LTL^T$ decomposition using a different order of operations in $n^3/3$ flops (see Rozložník et al. [2011] for a block version of Aasen's algorithm), as does the Bunch-Kaufmann algorithm [Bunch and Kaufman 1977] for computing a (complete) $LDL^T$ decomposition. Both algorithms can be generalized to the skew-symmetric case. Given the fact that computing the Pfaffian requires less information than a full tridiagonalization, it might seem feasible to compute the Pfaffian in $n^3/6$ flops. However, neither Aasen's algorithm (which is based on the fact that $TL^T$ is upper Hessenberg and hence $T$ fully tridiagonal), nor the Bunch-Kaufmann algorithm (which relies on the block-diagonal structure of $D$) are easily amended to compute a suitable partial factorization. Thus, for computing the Pfaffian, the Parlett-Reid algorithm is competitive. It remains an open question if it is possible to compute the Pfaffian of a dense skew-symmetric matrix in less than $n^3/3$ flops.

### 2.3. Tridiagonalization of Dense Matrices with Householder Reflections

Dense symmetric or Hermitian matrices are commonly reduced to tridiagonal form by Householder transformations [Golub and Van Loan 1996], and we adopt this approach to the skew-symmetric case here.

An order $m$ *Householder* transformation $H$ is a matrix of the form

$$H = 1 - \tau \mathbf{v}\mathbf{v}^\dagger$$

where $\tau \in \mathbb{C}$ and $\mathbf{v} \in \mathbb{C}^m$ chosen such that

$$H\mathbf{x} = \alpha ||x||_2 \mathbf{e}_1^{(m)}$$

for a given $\mathbf{x} \in \mathbb{C}^m$. Here $|| \cdot ||_2$ denotes the norm of a vector, $\mathbf{e}_1^{(m)}$ is the first unit vector in $\mathbb{C}^m$ and $\alpha \in \mathbb{C}$. For example, $\alpha = -e^{i\phi}$, if one chooses $\mathbf{v} = \mathbf{x} + e^{i\phi}||x||_2\mathbf{e}_1^{(m)}$ when $x_1 = e^{i\phi}|x_1|$, but there is a certain degree of freedom in choosing the Householder vector $\mathbf{v}$ which can be exploited to maximize stability (for an overview, see [Lehoucq 1995]). Note that $H$ is unitary (though not necessarily Hermitian) and can also be numerically calculated such that it is unitary up to machine precision [Golub and Van Loan 1996].

Householder transformations can thus be applied to a matrix to zero all the elements of a column (or row) below a chosen point, just as Gauss transformations, but without the need for pivoting. As a consequence, the structure of the tridiagonalization procedure is analogous to the $LTL^T$ decomposition. Assume that after step $k-1$ the matrix $H_{k-1} \dots H_1 A H_1^T \dots H_{k-1}^T$ is already tridiagonal in the first $k-1$ columns and rows and partitioned as defined in Equation (15). Then an order $n-k$ Householder matrix $\tilde{H}_k$ is chosen such that $\tilde{H}_k A_{32} \propto \mathbf{e}_1^{(n-k)}$ and the full transformation is set to $H_k = \mathrm{diag}(E_k, \tilde{H}_k)$. Writing $\tilde{H}_k = 1 - \tau \mathbf{v}\mathbf{v}^\dagger$ and defining $\mathbf{w} = \tau A_{33}\mathbf{v}^*$ we find

$$\tilde{H}_k A_{33} \tilde{H}_k^T = A_{33} + \mathbf{v}\mathbf{w}^T - \mathbf{w}\mathbf{v}^T . \tag{20}$$

The main difference to the $LTL^T$ decomposition is the fact that the computation of $\mathbf{w}$ now involves a full matrix-vector multiplication. Hence, the total computational cost of the outer product update in Equation (20) is $4(n-k)^2$ flops. The structure of a Householder tridiagonalization step is also shown schematically in Figure 1. The outer product updates of Equation (20) can be rearranged to increase the fraction of level-3 matrix operations. The block version of the Householder algorithm is detailed in Appendix 2.

Complete tridiagonalization with Householder matrices requires in total $4n^3/3$ flops. This can be reduced to $2n^3/3$ for computing the Pfaffian by skipping every other row/column elimination to compute only a partial tridiagonal form.

For the computation of the Pfaffian we also need to compute the determinant of the transformation matrix $Q = H_1^\dagger H_2^\dagger \dots H_{n-2}^\dagger$. The determinant of a single Householder transformation $H^\dagger = 1 - \tau^* \mathbf{v}\mathbf{v}^\dagger$ is given as

$$\det(H^\dagger) = 1 - \tau^* \mathbf{v}^\dagger \mathbf{v} . \tag{21}$$

For the particular choice $\tau = 2/\mathbf{v}^\dagger\mathbf{v}$, $\det(H) = \det(H^\dagger) = -1$, i.e. $P$ is a reflection, but other choices of $\tau$ are equally viable. In particular, if the matrix is already tridiagonal in certain column and row (which can happen frequently for very structured matrices), it is advantageous to use $H = E_n$. Moreover, any complex skew-symmetric matrix may be reduced to a purely real tridiagonal matrix using appropriate Householder transformations with complex $\tau$ [Lehoucq 1995]. Because of this, the determinant of each Householder reflection must be computed separately. The task of computing $\det(Q)$ still only scales as $\propto n^2$ and is thus negligible compared to the tridiagonalization cost.

In summary, for computing the Pfaffian Householder tridiagonalization is twice as costly as the Parlett-Reid algorithm and thus usually not competitive. It has however a right on its own given its connection to computing the canonical form of a skew-symmetric matrix.

### 2.4. Tridiagonalization of Band Matrices with Givens Rotations

The dense algorithms of the previous two sections are not easily amended to matrices with a finite band width. In the case of the Parlett-Reid algorithm, the symmetric pivoting can lead to an uncontrolled growth of the band width depending on the details of the matrix. In the Householder tridiagonalization, the outer product matrix update always introduces values outside the band, leading to a fast-growing band width.

For symmetric matrices, $LTL^T$ or $LDL^T$ decomposition algorithms respecting the band width have only been introduced recently [Irony and Toledo 2006; Kaufman 2007]. In contrast, banded tridiagonalization with unitary transformations is well established for symmetric matrices, and we adopt this approach for the skew-symmetric case that follows.

Instead of zeroing a whole column or row as is done in the Householder approach, for banded matrices it is of advantage to use a more selective approach. The method of choice for this case in the symmetric or Hermitian case are Givens rotations [Schwarz 1968], and we will extend this approach to th skew-symmetric case. A *Givens* rotation $G_{i,j}$ is a modification of the identity matrix that is only different in the $i$th and $j$th row and column. It is defined as

$$G_{i,j} = \begin{pmatrix} 1 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & & \vdots & & \vdots \\ 0 & \cdots & c & \cdots & s & \cdots & 0 \\ \vdots & & \vdots & \ddots & \vdots & & \vdots \\ 0 & \cdots & -s^* & \cdots & c & \cdots & 0 \\ \vdots & & \vdots & & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & 0 & \cdots & 1 \end{pmatrix} \begin{matrix} \\ \\ i \\ \\ j \\ \\ \\ \end{matrix} \tag{22}$$

with $c \in \mathbb{R}$, $s \in \mathbb{C}$ and $c^2 + |s|^2 = 1$ and thus clearly unitary. Choosing $c$ and $s$ such that

$$\begin{pmatrix} c & s \\ -s^* & c \end{pmatrix} \begin{pmatrix} x_i \\ x_j \end{pmatrix} = \begin{pmatrix} \tilde{x}_i \\ 0 \end{pmatrix}, \tag{23}$$

it is possible to selectively zero one element of a vector. Again, a Givens rotation can be computed numerically such that it is orthogonal up to machine precision.

A banded skew-symmetric matrix can be brought into tridiagonal form by Givens rotations of the form $G_{i,i+1}$. The structure induced in the process of applying $G_{i,i+1}$ from the left and right is shown schematically in Figure 2. Applying a Givens rotation $G_{i,i+1}$ ($G_{i,i+1}^T$) from the left (right) only modifies the $i$th and $(i+1)$th rows (columns). Due to the skew-symmetry, if $G_{i,i+1}$ zeroes the $(i+1)$ entry in column $j$, $G_{i,i+1}^T$ zeroes the $(i+1)$ entry in row $j$. Furthermore, each Givens rotation only introduces at most one additional nonzero entry outside the band in a row and column $k > i+1$. This nonzero entry can thus be moved further down the band by a sequence of Givens transformations until it is "chased" beyond the end of the matrix.

$$\begin{pmatrix}
0 & \times & \times & \boxdot & & & & & \\
\times & 0 & \times & \times & \times & & & & \\
\times & \times & 0 & \times & \times & \times & \boxtimes & & \\
\boxdot & \times & \times & 0 & \times & \times & \times & & \\
 & \times & \times & \times & 0 & \times & \times & \times & \\
 & & \times & \times & \times & 0 & \times & \times & \\
 & & \boxtimes & \times & \times & \times & 0 & \times & \\
 & & & & \times & \times & \times & 0 & \\
 & & & & & & & & \ddots
\end{pmatrix}$$

Fig. 2. Example for the structure induced by applying a Givens rotation $G_{2,3}$ to a skew-symmetric, banded matrix from the left and right: $\times$ denotes nonzero entries, $\boxdot$ the entry that is eliminated by the Givens transformation, and $\boxtimes$ the entries that are introduced outside band (fill-in). The Givens rotation only affect the second and third row and column (marked by frames).

The structure of the skew-symmetric tridiagonalization routine is thus identical to the symmetric or Hermitian case. The main difference is in the update of the diagonal $2 \times 2$-block that is affected by both Givens rotations from left and right: Due to the skew-symmetry, the diagonal blocks are invariant under these transformations,

$$\begin{pmatrix} c & s \\ -s^* & c \end{pmatrix} \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix} \begin{pmatrix} c & -s* \\ s & c \end{pmatrix} = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}. \tag{24}$$

Kaufman [1984, 2000] has presented a variant of the symmetric band matrix approach of Schwarz [1968] that permits working on more data in a single operation, which allows a more favorable memory access pattern. These modifications carry over unchanged to the skew-symmetric case.

The tridiagonalization of an $n \times n$ skew-symmetric matrix with bandwidth $b$ using Givens transformations scales as $\mathcal{O}(b\,n^2)$.

The determinant of any single Givens rotation $\det(G_{i,j}) = 1$, and thus the determinant of the full transformation $\det(Q) = 1$, too. In the complex case the resulting tridiagonal matrix can be chosen to be purely real, in this case the determinant of total unitary transformations (the Givens transformations and row/columns-scalings with a phase factor) obey $|\det(Q)| = 1$.

## 3. NOTES ON THE IMPLEMENTATION

### 3.1. Fortran

We have implemented the algorithms described in this manuscript as a comprehensive set of Fortran routines for real and complex variables as well as single and double precision. Because of the conceptional similarity of the skew-symmetric problem to the symmetric and Hermitian problem, these routines are designed analogous to the corresponding symmetric and Hermitian counterparts in LAPACK. Moreover, our implementation also makes use of the LAPACK framework for computing, applying, and accumulating Householder and Givens transformations, which was designed for numerical stability and which is available in an optimized form for any common computer architecture.

Dense skew-symmetric matrices are stored as ordinary two-dimensional Fortran matrices, but only the strictly lower or upper triangle needs to be set (for differences in the implementation between lower and upper triangular storage see Appendix 3). For banded skew-symmetric matrices, only the strictly upper or lower diagonals are stored in a `K` $\times$ `N` array `AB`, where `K` is the number of non-zero off-diagonals and `N`

Table I.

Overview of the computational routines in the Fortran implementation. In the Fortran77 interface the routine name must be preceded by either S (single precision), D (double precision), C (complex single precision), or Z (complex double precision)

| | |
|---|---|
| SKTRF | Skew-symmetric tridiagonal decomposition of a dense matrix using the block Parlett-Reid algorithm. |
| SKTF2 | Skew-symmetric tridiagonal decomposition of a dense matrix using the Parlett-Reid algorithm (unblocked version). |
| SKTRD | Skew-symmetric tridiagonalization of a dense matrix using block Householder reflections. |
| SKTD2 | Skew-symmetric tridiagonalization of a dense matrix using Householder reflections (unblocked version). |
| SKPFA | Compute the Pfaffian of a dense skew-symmetric matrix (makes use of either SKTRD or SKTRF). |
| SKPF10 | Compute the Pfaffian of a dense skew-symmetric matrix (makes use of either SKTRD or SKTRF). The result is returned as $a \times 10^b$ to avoid over- or underflow. |
| SKBTRD | Skew-symmetric tridiagonalization of a banded matrix using Givens rotations. |
| SKBPFA | Compute the Pfaffian of a banded skew-symmetric matrix (makes use of SKBTRD). |
| SKPF10 | Compute the Pfaffian of a banded skew-symmetric matrix (makes use of SKBTRD).The result is returned as $a \times 10^b$ to avoid over- or underflow. |

the size of the matrix. The j-th column of the matrix $A$ is stored in the j-th column of AB as

— AB(K + 1 + i − j, j) = $A_{i,j}$ for max(1, j − K) <= i <= j, if the upper triangle is stored,
— AB(1 + i − j, j) = $A_{i,j}$ for j <= i <= min(N, j + kd), if the lower triangle is stored.

Note that in this scheme, also the zero diagonal is explicitly stored. This was done to keep the design identical to the storage scheme of symmetric and Hermitian band matrices in LAPACK.

Our library includes stand-alone routines for the tridiagonalization of a skew-symmetric dense matrix (SKTRF and SKTF2 using the Parlett-Reid algorithm, SKTRD and SKTD2 using the Householder approach) and banded matrices (SKBTRD). We also include functions to compute the Pfaffian of a skew-symmetric dense (SKPFA and SKPF10) and banded matrices (SKBPFA and SKBPF10), which are based on the tridiagonalization functions. As the determinant, the Pfaffian of a large matrix is prone to floating point over or underflow. Because of that, we have included routines that return the Pfaffian in the form $a \times 10^b$, where $a$ is real or complex, and $b$ is always real and integer (SKPFA10 and SKBPF10). Both a Fortran95 and a Fortran77 interface, as well as a C interface are provided. In the Fortran77 version of the code the routine name is preceded by either S (single precision), D (double precision), C (complex single precision), or Z (complex double precision). The computational routines and their purposes are summarized in Table I.

The block versions of the algorithm have an internal parameter controlling the block size. By default, the routines use the same block sizes as their symmetric counterpart from the LAPACK library. However, this internal parameter may be changed by the user to optimize for a specific architecture.

Note that the algorithms overwrite the original matrix in the process and only require additional memory $\propto n$ (for a $n \times n$-matrix.)

Apart from the documentation here, all routines (including the auxiliary ones) are documented extensively in the respective files. Due to our routines using LAPACK and BLAS, both libraries must be also be linked.

### 3.2. Python, Matlab and Mathematica

While most compiled languages (including `C` and `C++`) are easily interfaced with a Fortran library, interpreted languages such as `Python` or programs such as Matlab or Mathematica require somewhat more effort. For this reason we have included stand-alone versions of the tridiagonalization of dense skew-symmetric matrices using Householder reflections implemented in `Python`, Matlab and Mathematica. Those implementations, being of course slower than the Fortran counterpart, are useful especially for situations where speed is not critical. They also make use of the fact that for computing the Pfaffian, only the odd rows and columns need to be tridiagonalized, but always work on the full matrix instead of a single triangle.

Again, more extensive documentation for both implementations may be found in the respective files.

## 4. EXAMPLES

### 4.1. Benchmarks

To demonstrate the effectiveness of our methods, we have performed benchmark computations of the Pfaffian of large, random matrices on various architectures. In Table II we compare our approach with the other available software that can also be used to calculate the Pfaffian in certain situations (see Section 1.3). For this benchmark we have compiled our Fortran implementation, and the implementations of Refs. [Ward and Gray 1978a] and [González-Ballestero et al. 2011] using the same compiler and compilation options, and chose a machine-optimized version of the LAPACK library [Anderson et al. 1999].

From the benchmark results we can see that the block approach is always faster than the unblocked version. The relative speed-up depends strongly on the architecture, but can reach up to 60%. We also observe that the relative speed-up of the Parlett-Reid algorithm is larger than that of the Householder tridiagonalization, reflecting the fact that the level-3 content of the former is larger (see Appendix 2)).

For the banded random matrices we observe that the Parlett-Reid algorithm performs surprisingly well. Although it is not designed to make use of the bandedness of the matrix, the implementation of the skew-symmetric outer product update takes into account zeros in the vectors of the update. The Householder tridiagonalization does not benefit as much, as for the matrices here the band width growth in the Householder approach is faster than in the Parlett-Reid algorithm. It should be stressed however that the performance of these algorithms in the banded case depends on the actual values of the matrix. For example, band width growth is stronger in the Parlett-Reid algorithm, if the largest entries sit at the edge of the band.

The specialized approach for banded matrices using Givens transformations is still slightly slower than the Parlett-Reid algorithm for the matrix sizes considered here. The main benefit of the specialized algorithm for Pfaffian calculations is hence its much lower memory requirement. In fact, typically the memory limits the matrix sizes that can be handled, not the computational time. The banded Givens-based approach however is considerably faster than the Householder tridiagonalization which makes it very attractive for computing the canonical form (or eigenvalues in the real case).

Comparing to other approaches, we observe that our routines are always faster, typically by a factor of about 10 or more. In particular in the real case, our specialized approach is considerably faster than using the real Hessenberg reduction,

Table II.

Benchmark comparison of the implementation of this work and other numerical approaches to compute the Pfaffian of a skew-symmetric matrix. The table shows the time needed to compute the Pfaffian for the various methods (time given in seconds) on two different architectures [(a) and (b)]. The first five columns of benchmark results correspond to the methods discussed in this work. `DGEHRD` is from LAPACK [Anderson et al. 1999], `TRIZD` from [Ward and Gray 1978a], and `PfaffianH` and `PfaffianF` from [González-Ballestero et al. 2011]. For the banded matrices, $k$ denotes the strictly upper or lower bandwidth, the full bandwidth is hence $2k + 1$.

| | Block Parlett-Reid | Regular Parlett-Reid | Block House-holder | Regular House-holder | Givens band matrix | `DGEHRD` | `TRIZD` | `PfaffianH` | `PfaffianF` |
|---|---|---|---|---|---|---|---|---|---|
| (a) benchmark for AMD Opteron 6174 2.2 Ghz | | | | | | | | | |
| real $3000 \times 3000$, dense | 5.1 | 5.9 | 9.4 | 10.5 | - | 24.7 | 383.4 | 54.5 | 120.8 |
| real $3000 \times 3000$, banded, $k = 100$ | 0.9 | 0.7 | 9.3 | 10.5 | 2.1 | 24.8 | 383.2 | 54.3 | 121.7 |
| complex $2000 \times 2000$, dense | 3.5 | 4.4 | 7.6 | 8.2 | - | - | - | 32.2 | 50.1 |
| complex $2000 \times 2000$, banded $k = 100$ | 0.8 | 0.7 | 7.6 | 8.1 | 2.0 | - | - | 31.9 | 50.2 |
| (b) benchmark for Intel Core 2 Duo E8135 2.66 Ghz | | | | | | | | | |
| real $3000 \times 3000$, dense | 3.5 | 8.3 | 8.4 | 12.4 | - | 30.7 | 105.3 | 76.3 | 49.4 |
| real $3000 \times 3000$, banded, $k = 100$ | 0.7 | 0.5 | 8.4 | 12.2 | 1.4 | 30.4 | 105.4 | 75.9 | 48.5 |
| complex $2000 \times 2000$, dense | 3.5 | 5.0 | 7.5 | 8.3 | - | - | - | 48.3 | 28.3 |
| complex $2000 \times 2000$, banded, $k = 100$ | 0.8 | 0.7 | 7.5 | 8.2 | 3.0 | - | - | 49.3 | 26.3 |

although we do not always reach the full speed-up of a factor of 10 that can be expected from the operation count, which is due to the optimization of the LAPACK library used. The implementation for real matrices of [Ward and Gray 1978a] is particularly slow as its memory access pattern is somewhat unfavorable for modern computer architectures.

### 4.2. Application: Topological Charge of a Disordered Nanowire

Finally, we apply our approach to computing the Pfaffian to a physical example, namely the numerical computation of the topological charge of a disordered nanowire.

A nanowire made out of a topological superconductor supports at its two ends Andreev bound states pinned at the Fermi energy [Kitaev 2001; Lutchyn et al. 2010; Oreg et al. 2010; Potter and Lee 2010; Wimmer et al. 2010]. Because of particle-hole symmetry, those states are Majorana fermions—particles that are their own antiparticle—and may allow for topologically protected quantum computing [Kitaev 2001]. In contrast, an ordinary (trivial) superconducting wire does not support such states. The recent proposal to realize a topological superconductor using ordinary semiconducting and superconducting materials [Lutchyn et al. 2010; Oreg et al. 2010; Potter

and Lee 2010] has stirred a lot of interest towards Majorana physics in condensed matter.

A topological charge $Q$ is a quantity that indicates whether a system is in the trivial or topological state, and hence signifies the absence or existence of Majorana bound states. A superconducting system exhibits particle-hole symmetry which allows the Hamiltonian to be written in the form [Kitaev 2001]

$$H = \frac{i}{4} \sum_{i,j} A_{ij} c_i c_j, \tag{25}$$

where $A$ is a skew-symmetric matrix, and $c_i$, $c_j$ Majorana operators with $c_i^\dagger = c_i$, $c_i^2 = 1$ and $c_i c_j + c_j c_i = \delta_{ij}$. We will further specialize to the case where particle-hole symmetry is the only remaining symmetry, that is, broken time-reversal and spin-rotation symmetry, which puts the system into class D of the general symmetry classification scheme [Evers and Mirlin 2008].

Kitaev has shown that for a translationally invariant wire,

$$Q = \operatorname{sign} \operatorname{Pf}[A(0)] \operatorname{Pf}[A(\pi/a_{\mathrm{uc}})] \tag{26}$$

is a topological charge that signifies the absence ($Q = 1$) or existence ($Q = -1$) of Majorana bound states at the ends. In this expression, $A(k)$ is the Hamiltonian in (Bloch) momentum space written in the Majorana basis. $A(k)$ is a matrix with a size corresponding to the size of the unit cell, the unit cell length is denoted by $a_{\mathrm{uc}}$. Note that the Pfaffian needs only be evaluated at two values of momenta which correspond to closing the unit cell with periodic ($k = 0$) and antiperiodic ($k = \pi/a_{\mathrm{uc}}$) boundary conditions.

For a clean system, the size of $A(k)$ is $\propto W$, where $W$ is the width of the wire, and Equation (26) has been used previously to compute the topological charge [Kitaev 2001; Lutchyn et al. 2010]. A disordered system can be considered (up to finite size effects, as follows) in Equation (26) as a large, disordered supercell repeated periodically. In this case, the size of $A(k)$ is $\propto WL$, where $L$ is the length of the supercell. This implies that $A(k)$ in a disordered system will be a very large matrix, and we are not aware of any application of Equation (26) for such a system. However, the sparse structure of $A$, in particular its bandedness, allows the application of the special algorithms developed in this work, and allows for the first time applying (26) to large, disordered systems.

Recently, an alternative definition of the topological charge for class D systems has been shown [Akhmerov et al. 2011]. In contrast to Equation (26) which is based on properties of the Hamiltonian, this alternative definition is based on transport properties:

$$Q = \operatorname{sign} \det r\,, \tag{27}$$

where $r$ is the reflection matrix. This definition is equally applicable to clean and disordered systems. In the following we show numerically the equivalence of the definitions (26) and (27).

For this we use the model of Refs. [Lutchyn et al. 2010; Oreg et al. 2010] with a normal state Rashba Hamiltonian

$$H_0 = \frac{\boldsymbol{p}^2}{2m_{\mathrm{eff}}} + U(\boldsymbol{r}) + \frac{\alpha_{\mathrm{so}}}{\hbar}(\sigma_x p_y - \sigma_y p_x) + \tfrac{1}{2} g_{\mathrm{eff}} \mu_B B \sigma_x\,, \tag{28}$$

where $m_{\mathrm{eff}}$ is the effective mass of the two-dimensional electron gas, $\alpha$ the Rashba spin-orbit coupling, and $g_{\mathrm{eff}}\mu_B B$ the Zeeman splitting due to an external magnetic field. Characteristic length and energy scales are $l_{\mathrm{so}} = \hbar^2/m_{\mathrm{eff}}\alpha_{\mathrm{so}}$ and $E_{\mathrm{so}} = m_{\mathrm{eff}}\alpha_{\mathrm{so}}^2/\hbar^2$. The electrons are then confined into a nanowire of width $W$ and length $L$ in the $x-y$-plane. For the numerical treatment, the Hamiltonian is discretized on a square grid

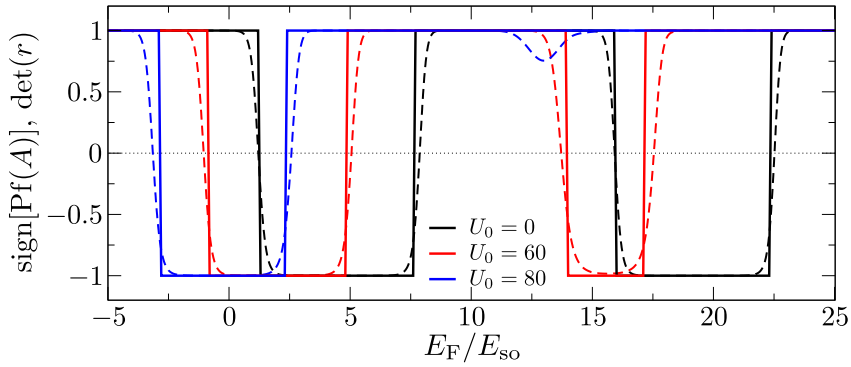


Fig. 3. Topological charge of a semiconductor nanowire in proximity to a superconductor for various disorder strengths as a function of the Fermi energy, computed from the Pfaffian of the Hamiltonian in the Majorana basis (26) (solid lines) and from the determinant of the reflection matrix (27) (dashed lines). Parameters of the calculation were $W = l_{so}$, $L = 10l_{so}$, $a = l_{so}/20$, $\Delta = 10E_{so}$, and $g_{eff}\mu_B B = 21$.

with lattice constant $a$ and thus represented by a matrix $H_{ij,\mu\nu}$, where $i$, $j$ denote lattice sites, and $\mu$, $\nu$ the spin degrees of freedom. Disorder is introduced as a random on-site potential taken from the uniform distribution $[-U_0, U_0]$. The Hamiltonian of the system in contact with a s-wave superconductor then reads

$$H = \sum_{i,j,\mu,\nu} H_{ij,\mu\nu} a^\dagger_{i,\sigma} a_{j,\nu} + \sum_i \Delta a^\dagger_{i,\uparrow} a^\dagger_{i,\downarrow} + \Delta a_{i,\downarrow} a_{i,\uparrow}\,, \tag{29}$$

where $\Delta$ is the proximity-induced pair potential. Defining Majorana operators as $c_{2i-1,\mu} = \frac{1}{\sqrt{2}}(a_{i,\mu} + a^\dagger_{i,\mu})$ and $c_{2i,\mu} = \frac{i}{\sqrt{2}}(a_{i,\mu} - a^\dagger_{i,\mu})$ we can transform Equation (29) into the form (25) with a skew-symmetric matrix $A$ (whose bandwidth is reduced for the numerics with a bandwidth reduction algorithm [Gibbs et al. 1976]).

In Figure 3 we show the numerical results for the topological charge as defined in Equations (26) and (27). For the computation of the Pfaffian in (26) we apply the Givens based method from Section 2.4, for computing the reflection matrix the numerical method of Wimmer and Richter [2009].

In all cases, clean and disordered, both definitions of $Q$ agree very well. In particular, both definitions predict a vanishing of the topological phase for the largest disorder in the region $10 < E_F/E_{so} < 25$. There are only very small differences in the exact value of $E_F$ where the topological charges change sign. These differences can be explained by finite size effects: At these points the bulk of the nanowire has a significant conductance ($|\det r| \ll 1$) which in turn means that the different geometry of Equations (26) (infinite repetition of a supercell) and (27) (single supercell connected to metallic leads) matter. In contrast, in the regime where the bulk of the wire is fully insulating ($|\det r| \approx 1$), both definitions agree fully.

The algorithmic developments in this work have allowed to evaluate Equation (26) for a fairly large disordered system. The bandwidth of the respective skew-symmetric matrix $A$ scales $\propto W$, and hence the cost of tridiagonalization as $\propto W^3L^2$. In contrast, the definition of the topological charge from transport properties (27) scales $\propto W^3L$ [Wimmer and Richter 2009]. Hence, from a computational viewpoint, Equation (27) is more favorable. It is thus reassuring that our numerical experiments showed the equivalence of both definitions.

## 5. CONCLUSIONS

We have shown that both the computation of the Pfaffian and the canonical form of a skew-symmetric matrix can be solved easily once the matrix is reduced to skew-symmetric (partial) tridiagonal form. To find this form, we have then developed tridiagonalization algorithms based on Gauss transformations, using a skew-symmetric, blocked version of the Parlett-Reid algorithm, and based on unitary transformations, using block Householder reflections and Givens rotations, applicable to dense and skew-symmetric matrices, respectively. These algorithms have been implemented in a comprehensive numerical library, and their performance has been proven to be superior to other approaches in benchmark calculations. Finally, we have applied our numerical method for computing the Pfaffian to the problem of computing the topological charge of a disordered nanowire, showing numerically the equivalence of different methods based on the Hamiltonian or the scattering matrix of the system.

## ACKNOWLEDGMENTS

The author acknowledges stimulating discussions with F. Hassler, I. C. Fulga, A. R. Akhmerov, C. W. Groth and C. W. J. Beenakker.

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Received February 2011; accepted July 2011